

Beer & Cask Ale - Kids

UK Family Tour

The Living Beer

Here is a strange question. Did you know that some drinks are alive? Not pretend alive. Actually alive, full of billions of tiny living creatures, right up until the moment someone drinks them.

Beer is one of the oldest drinks people have ever made. In a cave near the sea in Israel, scientists found the leftovers of brewing from thirteen thousand years ago. That is older than farming. Older than the wheel. Older than writing. Some scientists think people may have started planting the very first fields of grain partly because they wanted more beer. Imagine that. Fields, farms, villages, maybe whole cities, all starting with a bubbly drink.

Now, beer is a grown-up drink, like whisky, and it is not for kids. But the way it is made is one of the best science stories in Britain, because the maker of beer is a tiny living creature called yeast, the exact same creature that makes bread rise. Beer is basically bread's fizzy cousin. And in most of the world, beer gets filtered and zapped and sealed in cans, so nothing in it is alive any more. But here in Britain there is a special old kind called cask ale that stays alive all the way to the pub. It sits in a barrel in the cellar, a cool room under the floor, still bubbling away, and it gets pulled up through a pipe with a big lever on the bar, like a water pump from the olden days. Look for a row of tall handles standing on the bar. That is the clue.

In the next stories I'll show you the burping creature that makes it all work, the four friends who rescued this beer when it nearly disappeared, the days when this whole city smelled of warm toast, how to be a pub sign detective, and a beer that sailed all the way to India. Ready? Let's meet the yeast.

The Tiny Creature That Burps

Meet yeast. It is a living thing so small you could fit thousands of them on this full stop, and it has one favourite food in the whole world. Sugar. When yeast finds sugar, it gobbles it up, and then, there is no polite way to say this, it burps. Billions and billions of tiny burps of gas. Every bubble in bread, every bubble in real beer, is a yeast burp. You have been eating yeast burps your whole life, every time you eat a slice of toast. You're welcome.

So how do you make beer? You start with barley, a grain like wheat. The barley gets soaked in water and tricked into starting to sprout, as if spring has come, because sprouting fills the grain with sugar. Then it's dried and toasted. Now it's called malt, and it smells wonderful, like biscuits. The brewer grinds it up and stirs it into a giant pot of hot water, like making the world's biggest cup of barley tea. The sweet liquid that comes out even tastes like the milk left at the bottom of your cereal bowl. Brewers give it a funny old name, the wort. Say it like "wert."

Next the brewer boils the wort and throws in hops. A hop is a flower from a climbing plant, a little green papery cone, like a pinecone made of tissue paper, and it tastes amazingly bitter. That sounds like a mistake, but it's like putting lemon in something sweet. It balances the taste, and it helps stop the beer going bad.

Then comes the yeast's big moment. The brewer pours the cooled sweet liquid into a huge tank and adds the yeast, and the feast begins. For days the whole tank froths and fizzes and heaves like a bubble bath gone wild, while billions of yeast eat the sugar and burp out bubbles. The burping also makes the part that turns it into a grown-up drink, the alcohol. In a bakery, the same creature does the same trick in bread dough, the burps puff the bread up, and the oven's heat cooks everything else away. Same tiny creature. Bread or beer just depends on what you do with its burps.

One more secret, and you might remember it from the whisky story. If you take this bubbly barley liquid before the hops go in, and boil it in one of those giant copper kettles to catch the strong part, you're on the way to making whisky. So beer is really whisky's big sibling. Beer came first, thousands of years first, and whisky is what happens when beer gets boiled down and put to bed in a barrel for years.

So next time you see bread rising, or spot those bubbles in a glass, you'll know the truth. Somewhere in there, a crowd of the tiniest pets in the world just finished their dinner, and they all burped at once.

The Great Beer Rescue

Remember how I said cask ale is alive? Let me show you what that really means. In most of the world, when beer leaves the factory, it gets strained until nothing is left in it, heated to zap every last yeast, and pumped full of gas from a cylinder, like a fizzy drink. It's finished. Nothing in the can is alive.

But cask ale leaves the brewery early, on purpose, with its yeast still in the barrel, still eating, still burping. The barrel goes down into the pub's cellar, the cool stone room under the floor, and there the beer finishes making itself. The person who looks after it is called the cellarman, and the job is a bit like keeping a very fussy pet. The barrel has to rest on its side for days without being jiggled. It gets a special wooden peg knocked into it, a soft one first so the beer can breathe out its bubbles, then a hard one to keep the fizz in. The cellar has to stay cool, like a cave, never warm and never freezing. And once a barrel is opened, the beer has to be drunk within a few days, just like fresh bread from a bakery goes stale, because fresh things don't wait.

When somebody orders this beer, no machine pushes it. The person at the bar grabs one of those tall handles and pulls, one, two, three long slow pulls, and each pull sucks beer up a pipe from the cellar below, like a well pump. That lever machine is called a beer engine, and it was invented more than two hundred years ago. It still works exactly the same today. Watch for it. It looks like rowing a boat, standing up.

Now for the rescue. About fifty years ago, this living beer nearly vanished. A few giant companies had bought up most of the pubs in Britain, and dead fizzy beer was easier for them, no cellar, no fuss, no going stale. One by one, the hand pumps were ripped off the bars. Living beer was disappearing from the country that invented it, and hardly anyone seemed to care.

Then, in 1971, four friends from the north of England went on holiday together to Ireland. One evening they sat in a tiny pub in a village by the sea, moaning about how boring beer had become back home. And instead of just moaning, they decided, half as a joke, to start a club to fight back. They called it the Campaign for Real Ale, CAMRA for short. Four people. No money, no office, no plan, really.

And it worked. It worked amazingly. Thousands of people joined, then tens of thousands. They wrote a guidebook to the pubs that still kept living beer, and people queued up to buy it. They held beer festivals in big halls. Newspapers wrote about them. And the giant companies, one by one, gave in and put the hand pumps back. People now call it one of the most successful campaigns ordinary customers have ever won, anywhere.

So every tall handle you spot on a bar this week is really a little trophy. It's there because four friends on holiday got annoyed about something they loved disappearing, and actually did something about it.

The City That Smelled of Toast

Sniff the air. Nothing unusual today, probably. But if you had stood on this street a hundred years ago, the whole city would have smelled of warm toast and porridge. A sweet, cosy, biscuity smell drifting over the rooftops. That was the smell of malt being cooked, because Edinburgh, believe it or not, was one of the biggest beer-making cities in the whole world. Everyone remembers the whisky. Almost everyone has forgotten the beer.

It started nearly nine hundred years ago with monks. A Scottish king built a big abbey called Holyrood at the bottom of the Royal Mile, and the monks there brewed beer, the way monks did all over Europe. They had picked a lucky spot without knowing it. Deep under Edinburgh, water trickles through old stone and comes up through wells, and that water has minerals in it that happen to be perfect for brewing. Brewers later worked out exactly where the magic water ran, in a curving band right under the city, and they gave it a wonderful name. The charmed circle. If your brewery stood on the charmed circle, your beer came out better, so breweries squeezed onto that line like players fighting for the best spot in a game.

By Victorian times, Edinburgh had about forty breweries. Forty! Steam puffing, barrels rolling and rumbling over the cobbles on horse-drawn carts, and that toasty smell everywhere, all the time. The beers even had funny names. Scottish brewers named their beers after prices. A barrel of ordinary beer cost sixty shillings, which was old Scottish and English money, so the beer was called Sixty Shilling. Stronger beer cost more, so it was called Seventy Shilling, or Eighty Shilling. It's as if you invented a lemonade and just named it Three Pounds Fifty. And people still order it. You can walk into a pub here today and hear a grown-up ask for "an eighty bob," a beer named after its Victorian price tag. The strongest, darkest kind of all got the best name, though. They called it a wee heavy, which is Scots for a small mighty thing.

Bit by bit, the breweries closed down. Companies gobbled up other companies, buildings were knocked down, and the toast smell faded from the air. The very last of the old giants was called the Caledonian Brewery, and it kept brewing the old way until just a few years ago, boiling its beer in huge open copper kettles with real flames underneath, the last brewery in all of Britain still doing that. Then, in 2022, the company that owned it shut it down, after a hundred and fifty three years, and its buildings are being turned into houses. The people who worked there were heartbroken, and lots of the city was too.

But don't be too sad, because the story has a twist. All over Edinburgh, new little breweries have been popping up, small ones, started by people who love the old beer, filling barrels just like the monks and the Victorians did. The city that smelled of toast is quietly warming up the toaster again.

Pub Sign Detectives

Time to learn a detective game you can play on any British street. It's called reading the pubs.

First, the word itself. Pub is short for public house, and it means exactly what it says. Long ago, a pub really was somebody's house, where the family opened their front room to visitors, cooked food, and served drinks, while they lived upstairs. That's why a proper pub still feels like a cosy living room, with a fireplace, squashy seats, and very often a dog snoozing on the floor. And here's something visitors never expect. British pubs are for families too, especially at lunchtime. Kids come in with their parents for fish and chips or a big Sunday roast, and order lemonade or ginger beer, which is a fizzy, spicy, completely kid-friendly drink that tingles in your nose. A pub at lunchtime is one of the cosiest places in Britain to be hungry.

Now, the signs. Look up outside any pub and you'll see a painted picture hanging over the street. A red lion. A white horse. A king's head. A coach and horses. Those pictures are there because of a rule from more than six hundred years ago that said alehouses must hang out a sign, and in those days most people couldn't read. Not the customers, not even some of the owners. So instead of words, pubs used pictures,

and everyone knew them by sight. Meet you at the sign of the lion. The pictures are still up there, all over Britain, a giant outdoor puzzle museum that most people walk right past. Your detective job this week is to spot as many as you can. Score double for any animal wearing a crown.

And now, the best pub story I know in Edinburgh. In a few days you're climbing Arthur's Seat, the sleeping volcano. On the far side of it sits a tiny old village called Duddingston, and in that village is a pub called the Sheep Heid Inn. Yes, that means sheep's head, and there's a reason. Long ago the sheep that grazed around Arthur's Seat ended up in Duddingston, and the villagers got famous for cooking, well, every single part. The story goes there's been an inn here since 1360, more than six hundred and sixty years, which would make it one of the oldest pubs in all of Scotland. The story also goes that a Scottish king liked stopping here so much that he gave the innkeeper a fancy little box made from a ram's head. Nobody can prove those stories, that's why they're stories, but here is the part that's absolutely real, and it's the best part. Out the back is a skittle alley, a wooden bowling lane from Victorian times, said to be the oldest one left in Scotland. Skittles is bowling's great-grandparent. You roll a heavy ball down old wooden boards and try to knock over nine wooden pins, and the crash echoes off the walls exactly like it did when your great-great-great-grandparents were kids. Families can book a lane and play while their lunch is cooking.

So this is the plan. Climb the volcano first. Then walk down the far side to the little village, find the sign with the sheep's head on it, and go knock some pins over in a bowling alley older than your whole family tree.

The Beer That Sailed to India

Two hundred and fifty years ago, if you wanted to get from Britain to India, there was only one way. A wooden sailing ship, creaking and rolling for four, five, even six months, all the way down around the bottom of Africa and across the Indian Ocean. No fridges. No freezers. Food and drink just had to survive, and lots of it didn't. Water in barrels went green and slimy. Milk was hopeless. And this is where beer enters the story of the sea.

People love to tell it like this. British beer kept spoiling on the long voyage, until one clever London brewer invented a special new beer, extra strong and stuffed with extra hops, because hops work like a preservative, the way vinegar keeps pickles safe. His beer arrived in India tasting wonderful, everyone cheered, and it got the name India Pale Ale. Great story. But beer detectives, real historians who dig through dusty old papers, checked, and the truth is sneakier. That strong, hoppy beer already existed in Britain, and it just happened to be brilliant at sea voyages. The famous brewer mostly won because his brewery sat right next to the docks where the India ships loaded up. Nobody invented it for India. India just gave it its name. So when you hear the legend, you can smile and say, the story goes.

Here's another ocean-sized surprise. In the olden days, ordinary weak beer, called small beer, was something almost everybody drank, even children, even at breakfast. Ew, right? But think about it. Their water often came from dirty rivers and wells that could make you really sick, while small beer had been boiled while it was being made, so it was often the safer choice. Kids in this city, hundreds of years ago, drinking small beer with their morning porridge. Aren't you glad someone invented taps?

Beer kept travelling. In Bavaria, in Germany, brewers stored their beer in freezing cold caves in the mountainsides, and down in that cold dark a different kind of yeast slowly took over, one that likes the cold and sinks to the bottom. It made a cleaner, crisper, lighter beer, and its name comes from the German word for a storeroom. Lager. Then a town called Pilsen made the first golden, sparkling, sunshine-coloured lager, and the world went bananas for it. Today, if you picture a glass of beer, yellow and fizzy, you're picturing lager. Nearly every beer on the planet is now its grandchild.

Britain, though, stubborn as ever, kept its old living cask ale, with the barrels in the cellar and the levers

on the bar. And guess what happened. Brewers in America got bored of boring beer, discovered the old British styles, and made India Pale Ale famous all over again, so the beer that once sailed away to India ended up sailing home. Scotland is full of new brewers making it too.

So this week, when you're in a pub having your chips and your ginger beer, look over at the bar and count the tall handles. Every fizzy yellow lager in the world came from a cave in Germany. But the beer coming up through those levers was alive in the cellar right under your shoes a minute ago. Of all the beers on Earth, that one travelled the shortest way, and that's exactly what makes it special.